

Math 526, S12

Quiz 12

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given a matrix  $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ .

(a) Find the eigenvalues  $\lambda$ 's and corresponding eigenvectors  $x$ 's of  $A$ .

$$\det(A - \lambda I) = 0 \Rightarrow \det \begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix} = 0$$

$$(1-\lambda)(4-\lambda) - 10 = 0$$

$$\lambda^2 - 5\lambda - 6 = 0$$

$$(\lambda - 6)(\lambda + 1) = 0$$

$$\Rightarrow \lambda_1 = 6, \lambda_2 = -1.$$

For  $\lambda = 6$ . Solve  $(A - 6I)x_1 = 0$ .

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow x_1 = c \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

For  $\lambda = -1$ , Solve  $(A + I)x_2 = 0$

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow x_2 = c \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(b) Find an eigenvalue diagonalization  $A = SAS^{-1}$  based on the results from (a).

$$\text{Let } S = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}, \text{ then } S^{-1} = \frac{1}{2(-1) - 5(1)} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$

$$SAS^{-1} = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -1 \\ 30 & 1 \end{bmatrix} \begin{bmatrix} 1/7 & 1/7 \\ 5/7 & -2/7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$